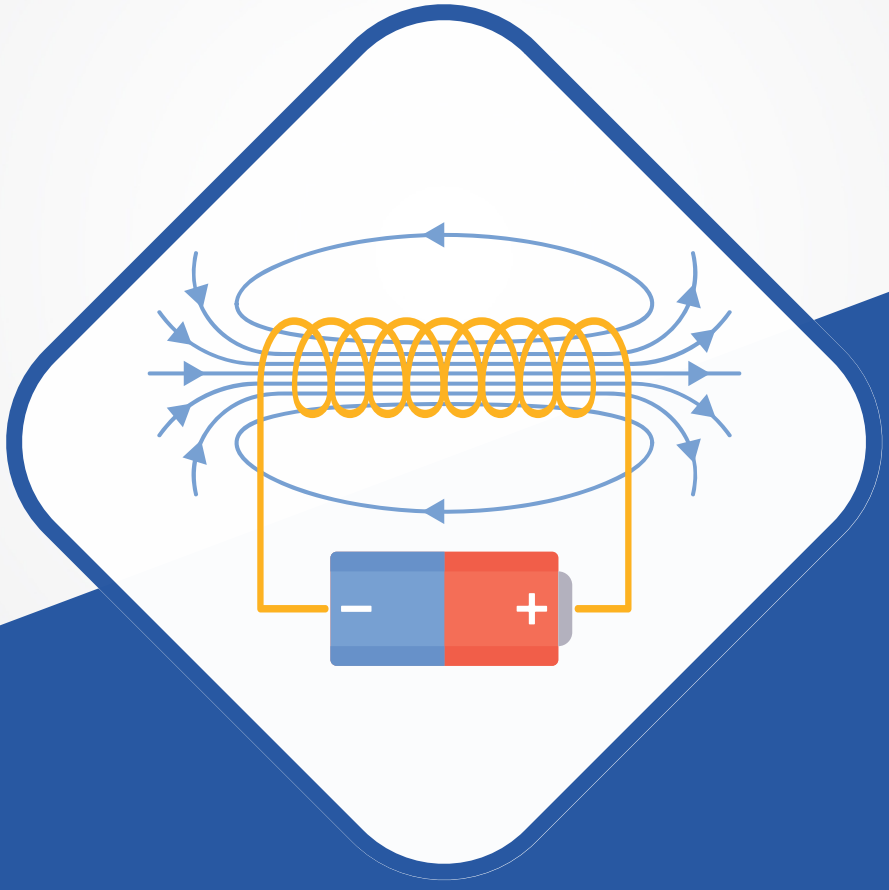
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KOTA (RAJASTHAN)
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Physics

ENTHUSIAST COURSE



STUDY MATERIAL

Electromagnetic Induction & Alternating Current

English Medium

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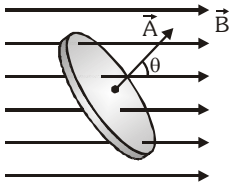
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ELECTROMAGNETIC INDUCTION & ALTERNATING CURRENT

KEY CONCEPT

1. MAGNETIC FLUX



The magnetic flux (ϕ) linked with a surface held in a magnetic field (B) is defined as the number of magnetic lines of force crossing that area (A). If θ is the angle between the direction of the field and normal to the area, (area vector) then $\phi = \vec{B}.\vec{A} = BA\cos\theta$

2. FLUX LINKAGE

If a coil has more than one turn, then the flux through the whole coil is the sum of the fluxes through the individual turns. If the magnetic field is uniform, the flux through one turn is $\phi = BA\cos\theta$. If the coil has N turns, the total flux linkage $\phi = NBA\cos\theta$

- Magnetic lines of force are imaginary, magnetic flux is a real scalar physical quantity with dimensions

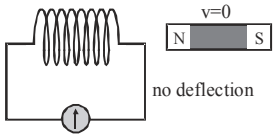
$$[\phi] = B \times \text{area} = \left[\frac{F}{IL}\right][L^2] \qquad \because\ B = \frac{F}{IL\sin\theta} \qquad [\because\ F = B\,I\,L\sin\theta]$$

$$\therefore \qquad [\phi] = \left[\frac{M\,L\,T^2}{A\,L}\right][L^2] = [M\,L^2\,T^{-2}\,A^{-1}]$$

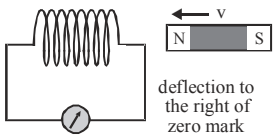
3. FARADAY'S EXPERIMENT

Faraday performed various experiments to discover and understand the phenomenon of electromagnetic induction. Some of them are :

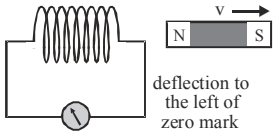
- When the magnet is held stationary anywhere near or inside the coil, the galvanometer does not show any deflection.



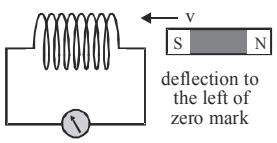
- When the N-pole of a strong bar magnet is moved towards the coil, the galvanometer shows a deflection right to the zero mark.



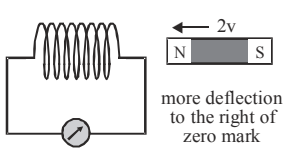
- When the N-pole of a strong bar magnet is moved away from the coil, the galvanometer shows a deflection left to the zero mark.



- If the above experiments are repeated by bringing the S-pole of the, magnet towards or away from the coil, the direction of current in the coil is opposite to that obtained in the case of N-pole.



- The deflection in galvanometer is more when the magnet moves faster and less when the magnet moves slow.



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4. FARADAY'S LAWS OF ELECTROMAGNETIC INDUCTION

Faraday's law of induction may be stated as follows:

The induced emf ε in a coil is proportional to the negative of the rate of change of magnetic flux:

$$\varepsilon = -\frac{d\Phi_B}{dt}$$

For a coil that consists of *N* loops, the total induced emf would be *N* times as large:

$$\varepsilon = -N\frac{d\Phi_B}{dt}$$

Thus, we see that an emf may be induced in the following ways:

(i) by varying the magnitude of $\vec{B}$ with time.

(ii) by varying the magnitude of $\vec{A}$, i.e., the area enclosed by the loop with time.

(iii) varying the angle between $\vec{B}$ and the area vector $\vec{A}$ with time.

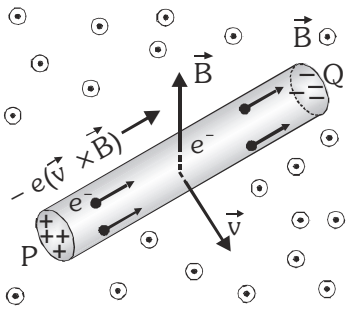
5. LENZ'S LAW

The direction of the induced current is determined by Lenz's law:

The induced current produces magnetic fields which tend to oppose the change in magnetic flux that induces such currents.

6. MOTIONAL EMF

A conductor PQ is placed in a uniform magnetic field B, directed normal to the plane of paper outwards. PQ is moved with a velocity v, the free electrons of PQ also move with the same velocity. The electrons experience a magnetic Lorentz force, $\vec{F}_m = (\vec{v}\times\vec{B})$. According to Fleming's left hand rule, this force acts in the direction PQ and hence the free electrons will move towards Q. A negative chagre accumulates at Q and a positive charge at P. An electric field E is setup in the conductor from P to Q. Force exerted by electric field on the free electrons is, $\vec{F}_e = e\vec{E}$



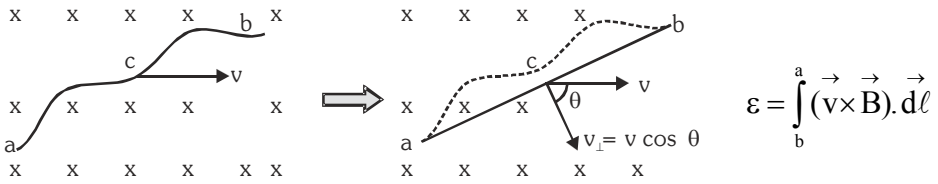
The accumulation of charge at the two ends continues till these two forces balance each other.

so $\vec{F}_m = -\vec{F}_e \Rightarrow e(\vec{v}\times\vec{B}) = -e\vec{E} \Rightarrow \vec{E} = -(\vec{v}\times\vec{B})$

For an element $\vec{d\ell}$ of conductor the contribution de to the emf is the magnitude $d\ell$ multiplied by the component of $\vec{v}\times\vec{B}$ parallel to $\vec{d\ell}$, that is $d\varepsilon = (\vec{v}\times\vec{B}).\vec{d\ell}$

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For any two points a and b the motional emf in the direction from b to a is,



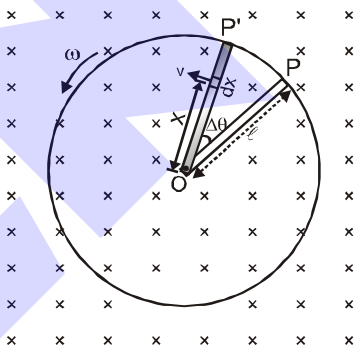
$$\varepsilon = \int_b^a (\vec{v}\times\vec{B}).\vec{d\ell}$$

Motional emf in wire acb in a uniform magnetic field is the motional emf in an imaginary wire ab. Thus, $e_{acb} = e_{ab}$ = (length of ab) ($v_\perp$) (B), $v_\perp$ = the component of velocity perpendicular to both $\vec{B}$ and ab. From right hand rule : b is at higher potential and a at lower potential. Hence, $V_{ba} = V_b - V_a$ = (ab) (v cosθ) (B)

7. INDUCED E.M.F. DUE TO ROTATION OF A CONDUCTOR ROD IN A UNIFORM MAGNETIC FIELD

$d\varepsilon = B\ v.\ dx$

Net induced e.m.f. across conducting rod

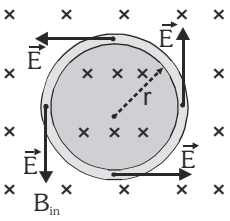


$$\varepsilon = \int d\varepsilon = \int_0^\ell B\omega\ x\ dx = \omega\ B\left(\frac{x^2}{2}\right)_0^\ell$$

or $$\varepsilon = \frac{1}{2}B\omega\ell^2$$

8. INDUCED ELECTRIC FIELD

When the magnetic field changes with time (let it increases with time) there is an induced electric field in the conductor caused by the changing magnetic flux.



$$\oint \vec{E}.\vec{d\ell} = e = -\frac{d\phi}{dt}$$

Important properties of induced electric field

(i) It is non conservative in nature. The line integral of $\vec{E}$ around a closed path is not zero. When a charge q goes once around the loop, the total work done on it by the electric field is equal to q times the emf. Hence

(ii) Being nonconservative field, so the concept of potential has no meaning for such a field.

(iii) This field is different from the conservative electrostatic field produced by stationary charges.

(iv) The relation $\vec{F} = q\vec{E}$ is still valid for this field.

Direction of induced electric field is the same as the direction of included current.

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9. GENERATORS

One of the most important applications of Faraday's law of induction is to generators and motors. A generator converts mechanical energy into electric energy, while a motor converts electrical energy into mechanical energy.

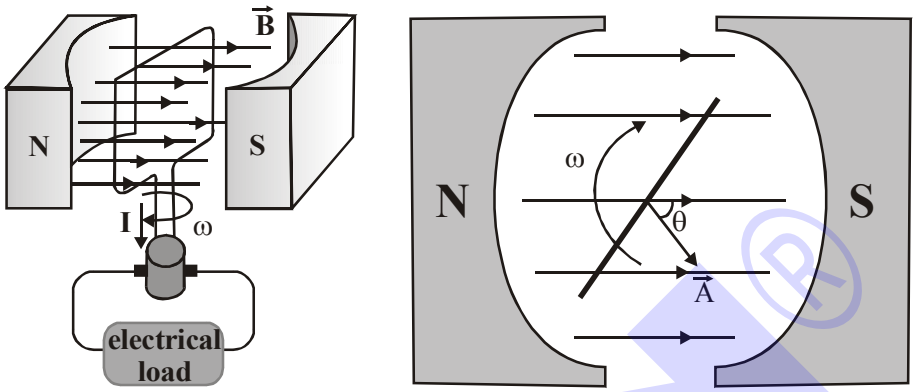


Figure : (a) A simple generator. (b) The rotating loop as seen from above.

Figure (a) is a simple illustration of a generator. It consists of an *N*-turn loop rotating in a magnetic field which is assumed to be uniform. The magnetic flux varies with time, thereby inducing an emf. From Figure (b), we see that the magnetic flux through the loop may be written as

$$\Phi_B = \vec{B} \cdot \vec{A} = BA\cos\theta = BA\cos\omega t$$

The rate of change of magnetic flux is

$$\frac{d\Phi_B}{dt} = -BA\,\omega\sin\omega t$$

Since there are N turns in the loop, the total induced emf across the two ends of the loop is

$$\varepsilon = -N\frac{d\Phi_B}{dt} = NB\,A\omega\sin\omega t$$

If we connect the generator to a circuit which has a resistance R, then the current generated in the circuit is given by $I = \frac{|\varepsilon|}{R} = \frac{NBA\omega}{R}\sin\omega t$

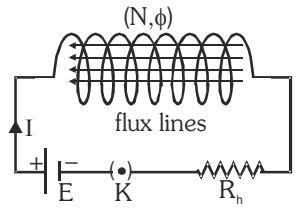
The current is an alternating current which oscillates in sign and has an amplitude $I_0 = NBA\omega / R$. The power delivered to this circuit is

$$P = I|\varepsilon| = \frac{(NBA\omega)^2}{R}\sin^2\omega t$$

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10. SELF INDUCTION

When the current through the coil changes, the magnetic flux linked with the coil also changes. Due to this change of flux a current induced in the coil itself according to lenz concept it opposes the change in magnetic flux. This phenomenon is called self induction.



total flux of coil ($N\phi$) $\propto$ current through the coil

$$N\phi \propto I \Rightarrow N\phi = LI \qquad L = \frac{N\phi}{I} = \frac{NBA}{I} = \frac{\phi_{total}}{I}$$

where L = coefficient of self inductance of coil

10.1 Self-Inductance of a Solenoid

The self-inductance of a solenoid with turns, length ℓ, and radius *NR* with a current *I* flowing through each turn, as shown in Figure.

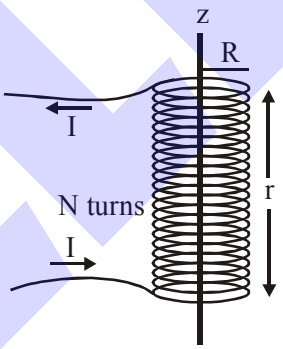


Figure : Solenoid

Ignoring edge effects and applying Ampere's law, the magnetic field inside a solenoid is given by Eq.:

$$\vec{B} = \frac{\mu_0 NI}{\ell}\hat{k} = \mu_0 nI \quad \hat{k}$$

where $n = N / \ell$ is the number of turns per unit length. The magnetic flux through each turn is

$$\Phi_B = BA = \mu_0 nI \,.\, (\pi R_2) = \mu_0 nI\pi R^2$$

Thus, the self-inductance is $L = \frac{N\Phi_B}{I} = \mu_0 n^2 \pi R^2 \ell$

We see that *L* depends only on the geometrical factors (*n*, *R* and ℓ) and is independent of the current *I*.

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10.2 Self-Inductance of a Toroid

Calculate the self-inductance of a toroid which consists of *N* turns and has a rectangular cross section, with inner radius *a*, outer radius *b* and height *h*, as shown in Figure (a).

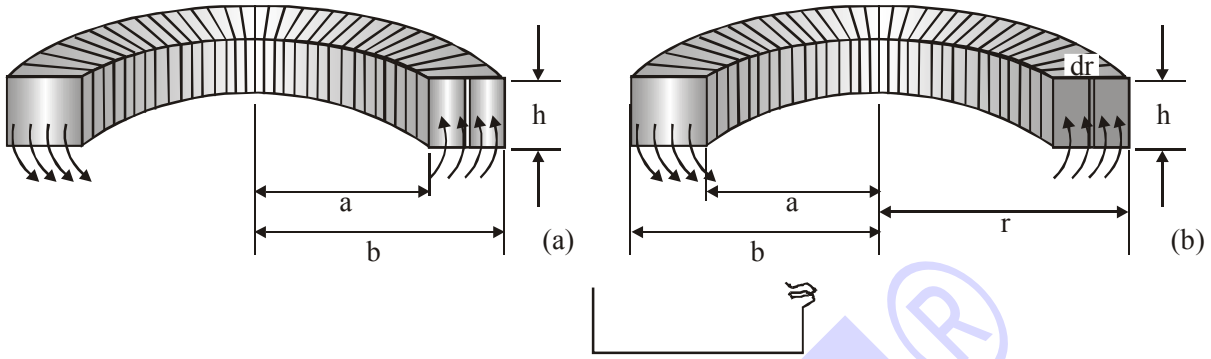


Figure : A toroid with *N* turns

According to Ampere's law discussed in section, the magnetic field is given by

$$\oint \vec{B}.d\vec{s}\ \oint Bds = B\oint ds = B\,(2\pi r) = \mu 0NI$$

or $$B = \frac{\mu_0 NI}{2\pi r}$$

The magnetic flux through one turn of the torid may be obtained by integrating over the rectangular cross section,k with dA = hdr as the differential area element (figure-b)

$$\Phi_B = \iint \vec{B}\ .\ d\vec{A} = \int_a^b \left(\frac{\mu_0 NI}{2\pi r}\right)\ hdr = \frac{\mu_0 NIh}{2\pi}\ln\left(\frac{b}{a}\right)$$

The total flux is $N\Phi_B$. Therefore, the self-inductance is

$$L = \frac{N\Phi_B}{I} = \frac{\mu_0 N^2 h}{2\pi}\ln\left(\frac{b}{a}\right)$$

Again, the self-inductance L depends only on the geometrical factors. Let's consider the situation where a >>b – a. In this limit, the logarithmic term in the equation above may be expanded as

$$\ln\left(\frac{b}{a}\right) = \ln\left(1+\frac{b-a}{a}\right) \approx \frac{b-a}{a}$$

and the self-inductance becomes

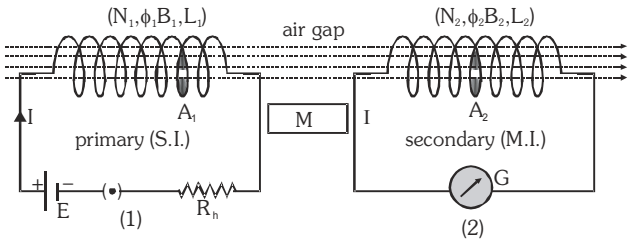
$$L \approx \frac{\mu_0 N^2 h}{2\pi}\cdot\frac{b-a}{a} = \frac{\mu_0 N^2 A}{2\pi a} = \frac{\mu_0 N^2 A}{\ell}$$

where A = h(b – a) is the cross-sectional area, and $\ell = 2\pi a$. We see that the self inductance of the torid in this limit has the same form as that of a solenoid.

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11. MUTUAL INDUCTION

Whenever the current passing through primary coil or circuit change then magnetic flux neighbouring secondary coil or circuit will also change. Acc. to Lenz law for opposition of flux change, so an emf induced in the neighbouring coil or circuit.



This phenomenon called as 'Mutual induction'. In case of mutual inductance for two coils situated close to each other, flux linked with the secondary due to current in primary.

$N_2 \phi_2 \propto I_1 \Rightarrow N_2 \phi_2 = MI_1$,

$M = \frac{N_2\phi_2}{I_1} = \frac{N_2B_1A_2}{I_1}$ where M : is coefficient of mutual induction.

- The mutual inductance does not depends upon current through the primary and it is constant for circuit system.

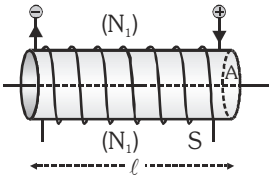
'M' depends on :

- Number of turns (N_1, N_2).
- Area of cross section.
- Distance between two coils (As d↓ = M↑).
- Cofficient of self inductance (L_1, L_2).
- Magnetic permeability of medium (μ_r).
- Orientation between two coils.

11.1 Different coefficient of mutual inductance

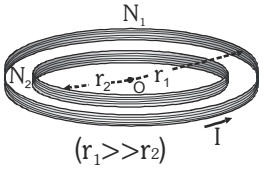
(a) Two co–axial solenoids

Coefficient of mutual inductance between two solenoids



$$M = \frac{N_2B_1A}{I_1} = \frac{N_2}{I_1}\left[\frac{\mu_0N_1I_1}{\ell}\right]A \Rightarrow M = \left[\frac{\mu_0N_1N_2A}{\ell}\right]$$

(b) Two plane concentric coils ($r_1 >> r_2$)



$$M = \frac{N_2B_1A_2}{I_1} \text{ where } B_1 = \frac{\mu_0N_1I_1}{2r_1},\ A_2 = \pi r_2^2$$

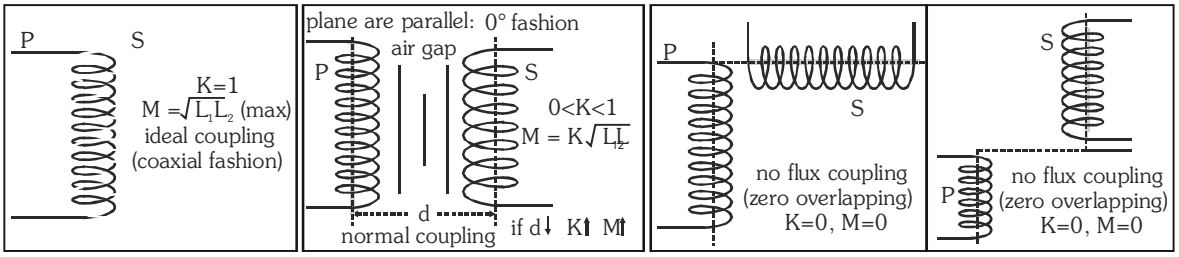
$$M = \frac{N_2}{I_1}\left[\frac{\mu_0N_1I_1}{2r_1}\right](\pi r_2^2) \Rightarrow M = \frac{\mu_0N_1N_2\pi r_2^2}{2r_1}$$

$M = K\sqrt{L_1L_2}$ here 'K' is coupling factor between two coils and its range $0 \le K \le 1$

- For ideal coupling $K_{max} = 1 \Rightarrow M_{max} = \sqrt{L_1L_2}$ (where M is geometrical mean of L_1 and L_2)
- For real coupling $(0 < K < 1)$ $M = K\sqrt{L_1L_2}$
- Value of coupling factor 'K' decided from fashion of coupling.

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- **Different fashion of coupling**



11.2 Inductance in series and parallel

Two coil are connected in series :

(a) If $M = 0$, $L = L_1 + L_2$ (b) If $M \neq 0$, $L = L_1 + L_2 \pm 2M$

Two coils are connected in parallel :

(a) If $M = 0$, $L = \dfrac{L_1 L_2}{L_1 + L_2}$ (b) If $M \neq 0$, $L = \dfrac{L_1 L_2 - M^2}{L_1 + L_2 \pm 2M}$

12. ENERGY STORED IN INDUCTOR

The energy of a capacitor is stored in the electric field between its plates. Similary, an inductor has the capability of storing energy in its magnetic field.An increasing current in an inductor causes an emf between its terminals.

Power P = The work done per unit time $= \dfrac{dW}{dt} = -ei = -\left[L\dfrac{di}{dt}\right]i = -L\,i\dfrac{di}{dt}$

so $\dfrac{dW}{dt} = -\dfrac{dU}{dt}$ $\therefore$ $\dfrac{dU}{dt} = Li\dfrac{di}{dt} \Rightarrow dU = Li\,di$

The total energy U supplied while the current increases from zero to final value i is,

$$U = L\int_0^I i\,di = \frac{1}{2}L(i^2)_0^I \quad \therefore U = \frac{1}{2}L\,I^2$$

the energy stored in the magnetic field of an inductor when a current I is $= \dfrac{1}{2}L\,I^2$.

The source of this energy is the external source of emf that supplies the current.

13. MAGNETIC ENERGY PER UNIT VOLUME OR ENERGY DENSITY

The energy is an inductor is actually stored in the magnetic field within the coil. For a long solenoid its magnetic field can be assumed completely within the solenoid.

The energy U stored in the solenoid when a current I is,

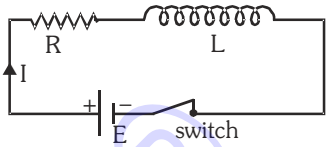
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$$U = \frac{1}{2} L I^2 = \frac{1}{2}(\mu_0 n^2 V) I^2 \qquad (L = \mu_0 n^2 V) \qquad (V = \text{Volume} = A\ell)$$

The energy per unit volume $u = \frac{U}{V} = \frac{1}{2}\mu_0 n^2 I^2 = \frac{(\mu_0 n I)^2}{2\mu_0} = \frac{B^2}{2\mu_0}$ $\quad (B = \mu_0 n I) \therefore u = \frac{1}{2}\frac{B^2}{\mu_0}$

14. R-L DC CIRCUIT

Current Growth



(i) emf equation $E = IR + L\frac{dI}{dt}$

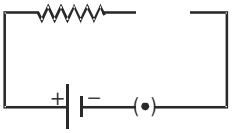
(ii) Current at any instant

When key is closed the current in circuit increases exponentially with respect to time. The current in circuit at any instant 't' given by $I = I_0\left[1 - e^{\frac{-t}{\tau}}\right]$

$t = 0$ (just after the closing of key) $\Rightarrow I = 0$

$t = \infty$ (some time after closing of key) $\Rightarrow I \to I_0$

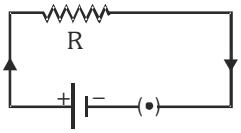
(iii) Just after the closing of the key inductance behaves like open circuit and current in circuit is zero.



Open circuit, $t = 0$, $I = 0$

Inductor provide infinite resistence

(iv) Some time after closing of the key inductance behaves like simple connecting wire (short circuit) and current in circuit is constant.



Short circuit, $t \to \infty$, $I \to I_0$,

Inductor provide zero resistence $I_0 = \frac{E}{R}$

(Final, steady, maximum or peak value of current) or ultimate current

Note : Peak value of current in circuit does not depends on self inductance of coil.

(v) Time constant of circuit (τ)

It is a time in which current increases up to 63% or 0.63 times of peak current value. $\tau = \frac{L}{R}$

(vi) Half life (T)

It is a time in which current increases upto 50% or 0.50 times of peak current value.

$$\Rightarrow \frac{I_0}{2} = I_0(1 - e^{-T/\tau}) \Rightarrow e^{-T/\tau} = \frac{1}{2}$$

$$T = 0.693\frac{L}{R}$$

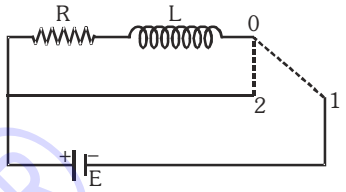
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(vii) Rate of growth of current at any instant

$$\left[\frac{dI}{dt}\right] = \frac{E}{L}(e^{-t/\tau}) \Rightarrow t = 0 \Rightarrow \left[\frac{dI}{dt}\right]_{max} = \frac{E}{L} \quad t = \infty \Rightarrow \left[\frac{dI}{dt}\right] \to 0$$

Note : Maximum or initial value of rate of growth of current does not depends upon resistance of coil.

Current Decay



(i) Emf equation $IR + L\frac{dI}{dt} = 0$

(ii) Current at any instant

Once current acquires its final max steady value, if suddenly switch is put off then current start decreasing exponentially wrt to time. At switch put off condition $t = 0$, $I = I_0$, source emf E is cut off from circuit $I = I_0(e^{-t/\tau})$

Just after opening of key $t = 0 \Rightarrow I = I_0 = \frac{E}{R}$

Some time after opening of key $t \to \infty \Rightarrow I \to 0$

(iii) Time constant (τ)

It is a time in which current decreases up to 37% or 0.37 times of peak current value.

(iv) Half life (T)

It is a time in which current decreases upto 50% or 0.50 times of peak current value.

(v) Rate of decay of current at any instant

$$\left[-\frac{dI}{dt}\right] = \left[\frac{E}{L}\right]e^{-t/\tau} \qquad t = 0, \left[-\frac{dI}{dt}\right]_{max.} = \frac{E}{L} \quad t \to \infty \Rightarrow \left[-\frac{dI}{dt}\right] \to 0$$

15. LC OSCILLATIONS

Consider an *LC* circuit in which a capacitor is connected to an inductor, as shown in Figure.

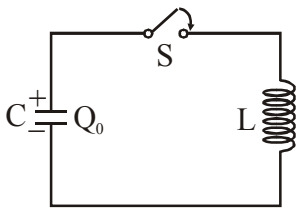


Figure : *LC* Circuit

Suppose the capacitor initially has charge Q_0. When the switch is closed, the capacitor begins to discharge and the electric energy is decreased. On the other hand, the current created from the discharging process generates magnetic energy which then gets stored in the inductor. In the absence of resistance, the total energy is transformed back and forth between the electric energy in the capacitor and the magnetic energy in the inductor. This phenomenon is called electromagnetic oscillation.

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The total energy in the *LC* circuit at some instant after closing the switch is

$$U = U_C + U_L = \frac{1}{2}\frac{Q^2}{C} + \frac{1}{2}LI^2$$

The fact that U remains constant implies that

$$\frac{dU}{dt} = \frac{d}{dt}\left(\frac{1}{2}\frac{Q^2}{C}+\frac{1}{2}LI^2\right) = \frac{Q}{C}\frac{dQ}{dt} + LI\frac{dI}{dt} = 0$$

$$\frac{Q}{C} + L\frac{d^2Q}{dt^2} = 0$$

Thus, the solutions for the charge and the current in our LC circuit are

$Q(t) = Q_0 \cos \omega_0 t$

and $I(t) = I_0 \sin w_0 t$

The time dependence of Q(t) and I(t) are depicted in figure.

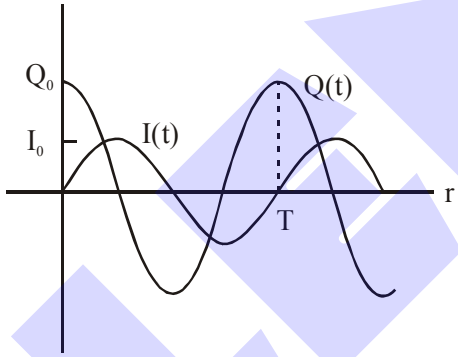


Figure : Charge and current in the *LC* circuit as a function of time

Using Eqs., we see that at any instant of time, the electric energy and the magnetic energies are given by

$$U_E = \frac{Q^2(t)}{2C} = \left(\frac{Q_0^2}{2C}\right)\cos^2 \omega_0 t$$

and $U_B = \frac{1}{2}LI^2(t) = \frac{LI_0^2}{2}\sin^2 \omega t = \frac{L(-\omega_0 Q_0)^2}{2}\sin^2 \omega_0 t = \left(\frac{Q_0^2}{2C}\right)\sin^2 \omega_0 t$

The electric and magnetic energy oscillation is illustrated in figure.

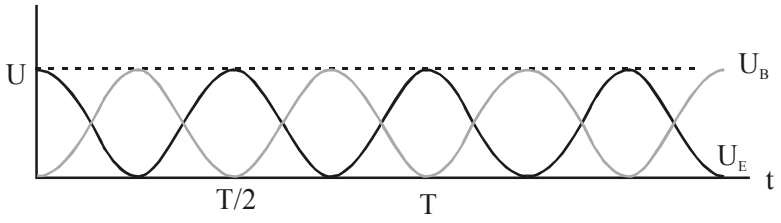


Figure : Electric and magnetic energy oscillations

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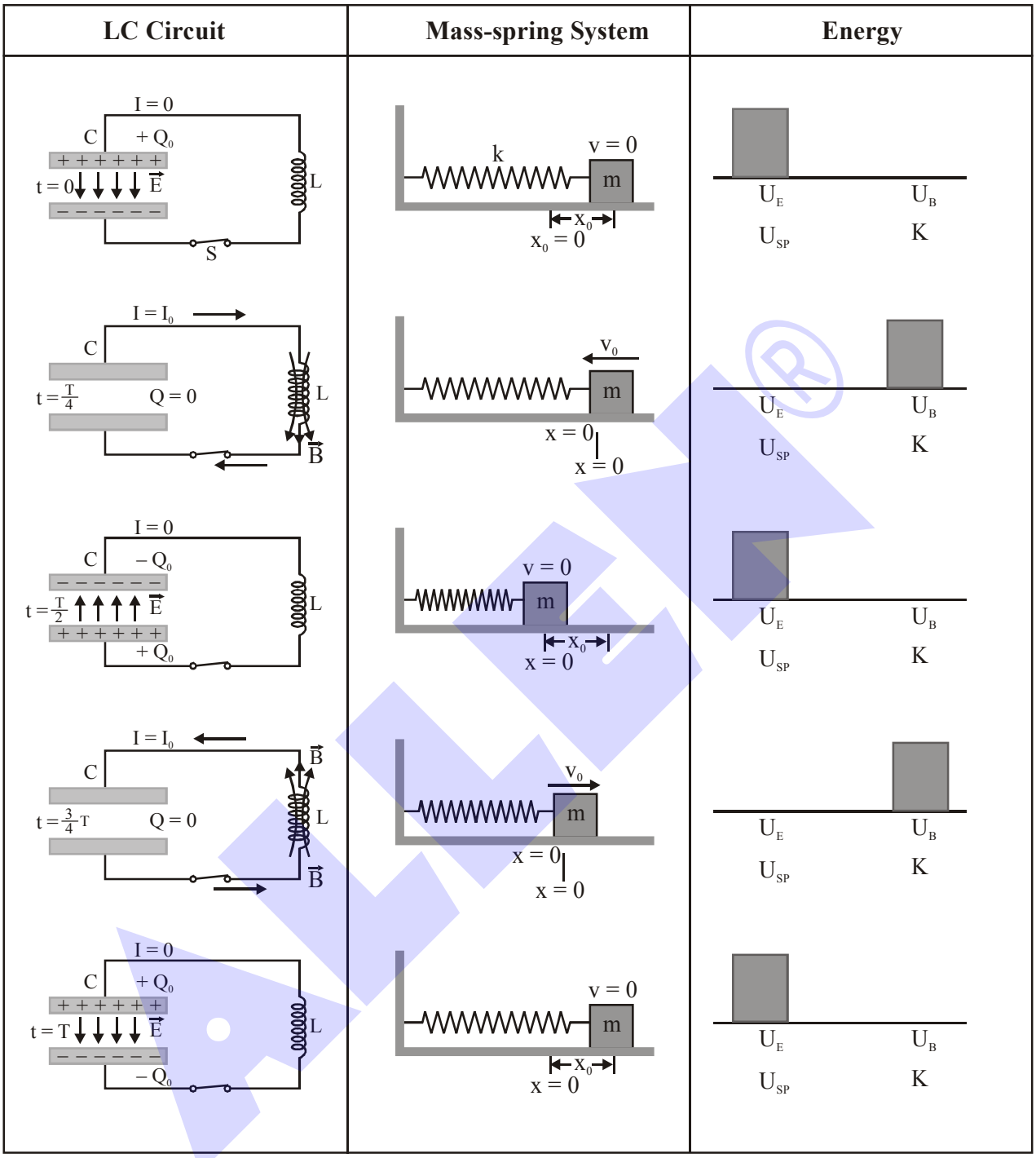


Figure : Energy oscillations in the *LC* Circuit and the mass-spring system

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ALTERNATING CURRENT

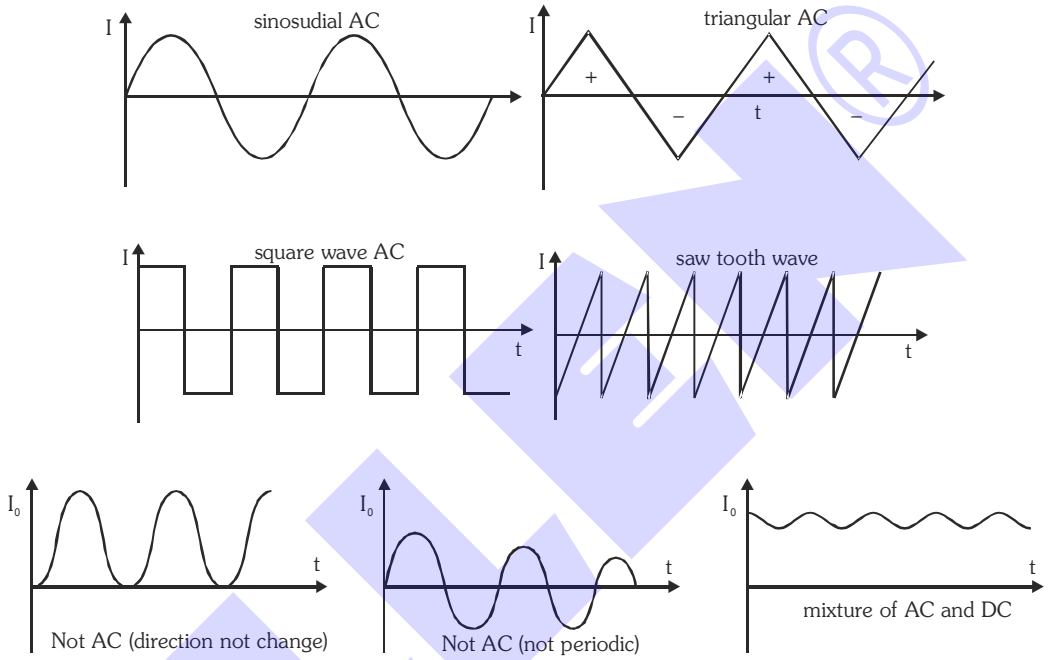
1. ALTERNATING CURRENT AND VOLTAGE

Voltage or current is said to be alternating if it is change continously in magnitude and perodically in direction with time.

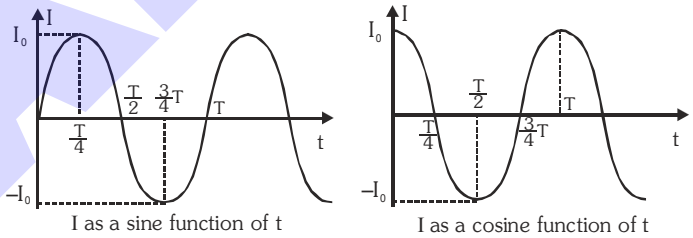
1.1 Condition required for current/ voltage to be Alternating

- Amplitude is constant
- Alternate half cycle is positive and half negative

The alternating current continuously varies in magnitude and periodically reverses its direction.



For a sine function $i = i_0 \sin(\omega t + \phi)$



1.2 Amplitude of ac

The maximum value of current in either direction is called peak value or the amplitude of current. It is represented by I_0. Peak to peak value = $2I_0$

1.3 Periodic Time

The time taken by alternating current to complete one cycle of variation is called periodic time or time period of the current.

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1.4 Frequency

The number of cycle completed by an alternating current in one second is called the frequency of the current.

UNIT : cycle/s ; (Hz)

In India : f = 50 Hz, supply voltage = 220 volt **In USA :** f = 60 Hz ,supply voltage = 110 volt

1.5 Average value or mean value

The mean value of A.C over any half cycle (either positive or negative) is that value of DC which would send same amount of charge through a circuit as is sent by the AC through same circuit in the same time.

$$\text{average value of current for half cycle } < I > = \frac{\int_0^{T/2} I\,dt}{\int_0^{T/2} dt}$$

Average value of $I = I_0 \sin \omega t$ over the positive half cycle :

$$I_{av} = \frac{\int_0^{\frac{T}{2}} I_0 \sin \omega t\, dt}{\int_0^{\frac{T}{2}} dt} = \frac{2I_0}{\omega T}\left[-\cos \omega t\right]_0^{\frac{T}{2}} = \frac{2I_0}{\pi}$$

$< \sin \theta > = < \sin 2\theta > = 0$
$< \cos \theta > = < \cos 2\theta > = 0$
$< \sin \theta \cos \theta > = 0$
$< \sin^2 \theta > = < \cos^2 \theta > = \frac{1}{2}$

- For symmetric AC, average value over full cycle = 0,
  Average value of sinusoidal AC

Full cycle	(+ve) half cycle	(–ve) half cycle
0	$\frac{2I_0}{\pi}$	$\frac{-2I_0}{\pi}$

As the average value of AC over a complete cycle is zero, it is always defined over a half cycle which must be either positive or negative

1.6 Root mean square (rms) value

It is value of DC which would produce same heat in given resistance in given time as is done by the alternating current when passed through the same resistance for the same time.

$$I_{rms} = \sqrt{\frac{\int_0^T I^2 dt}{\int_0^T dt}}$$ rms value = Virtual value = Apparent value

rms value of $I = I_0 \sin \omega t$:

$$I_{rms} = \sqrt{\frac{\int_0^T (I_0 \sin \omega t)^2 dt}{\int_0^T dt}} = \sqrt{\frac{I_0^2}{T}\int_0^T \sin^2 \omega t\, dt}$$

$$= I_0\sqrt{\frac{1}{T}\int_0^T \left[\frac{1-\cos 2\omega t}{2}\right] dt} = I_0\sqrt{\frac{1}{T}\left[\frac{t}{2} - \frac{\sin 2\omega t}{2\times 2\omega}\right]_0^T} = \frac{I_0}{\sqrt{2}}$$

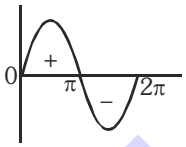
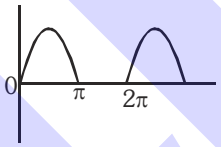
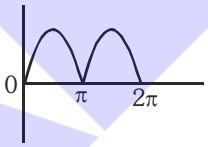
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- If nothing is mentioned then values printed in a.c circuit on electrical appliances, any given or unknown values, reading of AC meters are assumed to be RMS.

Current	Average	Peak	RMS	Angular fequency
$I_1 = I_0 \sin \omega t$	0	I_0	$\frac{I_0}{\sqrt{2}}$	ω
$I_2 = I_0 \sin \omega t \cos \omega t = \frac{I_0}{2} \sin 2\omega t$	0	$\frac{I_0}{2}$	$\frac{I_0}{2\sqrt{2}}$	2ω
$I_3 = I_0 \sin\omega t + I_0 \cos\omega t$	0	$\sqrt{2}\, I_0$	I_0	ω

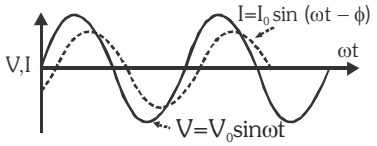
Some important wave forms and their rms and average value

Nature of wave form	Wave–form	RMS Value	Average or mean Value
Sinusoidal		$\frac{I_0}{\sqrt{2}} = 0.707\, I_0$	$\frac{2I_0}{\pi} = 0.637\, I_0$
Half wave rectifired		$\frac{I_0}{2} = 0.5\, I_0$	$\frac{I_0}{\pi} = 0.318\, I_0$
Full wave rectifired		$\frac{I_0}{\sqrt{2}} = 0.707\, I_0$	$\frac{2I_0}{\pi}$ $0.637\, I_0$

(a) V leads I or I lags V → It means, V reach maximum before I

Let if $V = V_0 \sin \omega t$ then $I = I_0 \sin (\omega t - \phi)$

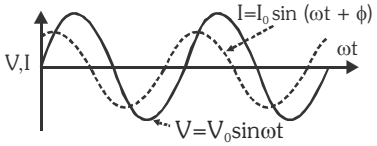
and if $V = V_0 \sin (\omega t + \phi)$ then $I = I_0 \sin \omega t$



(b) V lags I or I leads V → It means V reach maximum after I

Let if $V = V_0 \sin \omega t$ then $I = I_0 \sin (\omega t + \phi)$

and if $V = V_0 \sin (\omega t - \phi)$ then $I = I_0 \sin \omega t$



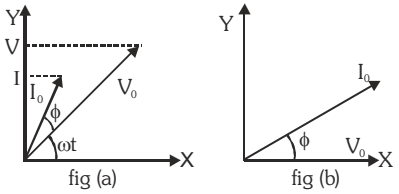
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2. PHASOR AND PHASOR DIAGRAM

A diagram representing alternating current and voltage (of same frequency) as vectors (phasor) with the phase angle between them is called phasor diagram.

Let $V = V_0 \sin \omega t$ and $I = I_0 \sin (\omega t + \phi)$

In figure (a) two arrows represents phasors. The length of phasors represents the maximum value of quantity. The projection of a phasor on y-axis represents the instantaneous value of quantity

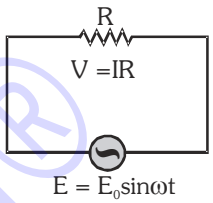


3. AC CIRCUIT CONTAINING PURE RESISTANCE

Let at any instant t the current in the circuit = I.

Potential difference across the resistance = I R.

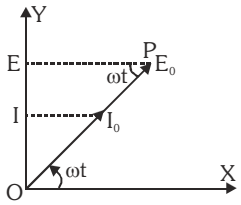
with the help of kirchoff's circuital law E – I R = 0



$\Rightarrow$ $E_0 \sin \omega t = I R$

$\Rightarrow$ $I = \frac{E_0}{R} \sin \omega t = I_0 \sin \omega t$ $\left(I_0 = \frac{E_0}{R} = \text{peak or maximum value of current}\right)$

i.e. $I = I_0 \sin\omega t$ and $E = E_0 \sin\omega t$.



Since $I_0 = \frac{E_0}{R}$,

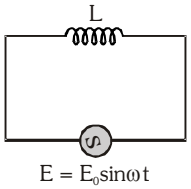
hence $\frac{I_0}{\sqrt{2}} = \frac{E_0}{R\sqrt{2}}$

$\Rightarrow I_{rms} = \frac{E_{rms}}{R}$

4. AC CIRCUIT CONTAINING PURE INDUCTANCE

A circuit containing a pure inductance L (having zero ohmic resistance) connected with a source of alternating emf.

Let the alternating e.m.f. $E = E_0 \sin \omega t$



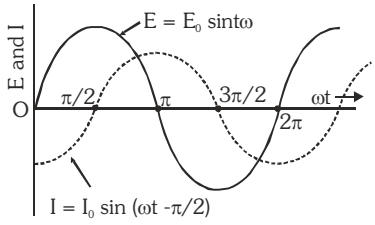
When a.c. flows through the circuit, emf induced across inductance $= -L\frac{dI}{dt}$

Negative sign indicates that induced emf acts in opposite direction to that of applied emf.

Because there is no other circuit element present in the circuit other then inductance so with the help of

Kirchoff's circuital law $E + \left(-L\frac{dI}{dt}\right) = 0 \Rightarrow E = L\frac{dI}{dt}$

so we get $I = \frac{E_0}{\omega L} \sin\left(\omega t - \frac{\pi}{2}\right)$

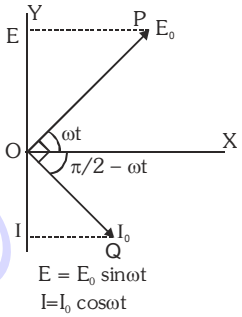


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Maximum current $I_0 = \frac{E_0}{\omega L} = \frac{E_0}{X_L}$, Hence, $I = I_0 \sin\left(\omega t - \frac{\pi}{2}\right)$

In a pure inductive circuit current always lags behind the emf by $\frac{\pi}{2}$.

or alternating emf leads the a. c. by a phase angle of $\frac{\pi}{2}$.

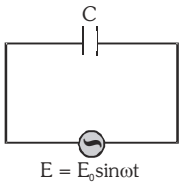


This non-resistive opposition to the flow of A.C. in a circuit is called the inductive reactance (X_L) of the circiut.

$X_L = \omega L = 2\pi f L$ where f = frequency of A.C.

5. AC CIRCUIT CONTAINING PURE CAPACITANCE

A circuit containing an ideal capacitor of capacitance C connected with a source of alternating emf as shown in fig. The alternating e.m.f. in the circuit $E = E_0 \sin \omega t$ When alternating e.m.f. is applied across the capacitor a similarly varying alternating current flows in the circuit.

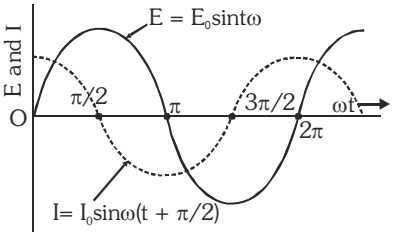


Instantaneous potential difference across the capacitor $E = \frac{q}{C}$

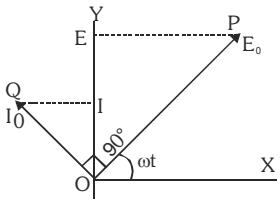
$\Rightarrow \quad q = C\,E \quad \Rightarrow q = CE_0 \sin \omega t$

The instantaneous value of current

$$I = \frac{dq}{dt} = \frac{d}{dt}\left(CE_0 \sin \omega t\right) = CE_0 \omega \cos \omega t$$



$\Rightarrow \quad I = \frac{E_0}{(1/\omega C)} \sin\left(\omega t + \frac{\pi}{2}\right) = I_0 \sin\left(\omega t + \frac{\pi}{2}\right)$ where $I_0 = \frac{E_0}{X_C}$

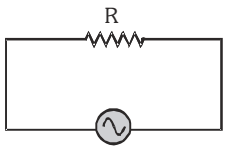
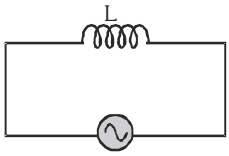
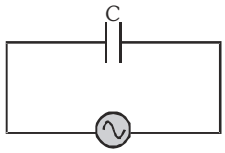
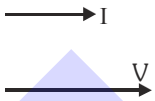
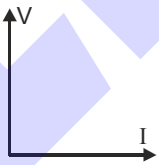
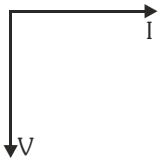
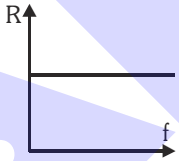
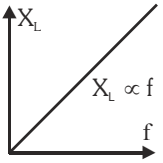
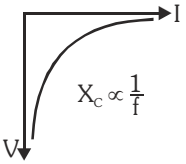


In a pure capacitive circuit, the current always leads the e.m.f. by a phase angle of π/2. The alternating emf lags behinds the alternating current by a phase angle of π/2.

This non-resistive opposition to the flow of A.C. in a pure capacitive circuit is known as capacitive reactance X_C. $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$

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Individual Components (R or L or C)

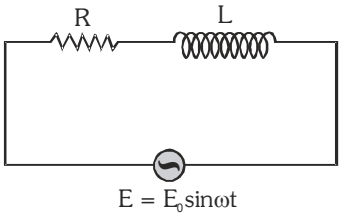
Term	R	L	C
Circuit			
Supply Voltage	$V = V_0 \sin \omega t$	$V = V_0 \sin \omega t$	$V = V_0 \sin \omega t$
Current	$I = I_0 \sin \omega t$	$I = I_0 \sin (\omega t - \frac{\pi}{2})$	$I = I_0 \sin (\omega t + \frac{\pi}{2})$
Peak Current	$I_0 = \frac{V_0}{R}$	$I_0 = \frac{V_0}{\omega L}$	$I_0 = \frac{V_0}{1/\omega C} = V_0 \omega C$
Impedance (Ω)	$\frac{V_0}{I_0} = R$	$\frac{V_0}{I_0} = \omega L = X_L$	$\frac{V_0}{I_0} = \frac{1}{\omega C} = X_C$
$Z = \frac{V_0}{I_0} = \frac{V_{rms}}{I_{rms}}$	R = Resistance	X_L =Inductive reactance.	X_C = Capacitive reactance.
Phase difference	zero (in same phase)	$+\frac{\pi}{2}$ (V leads I)	$-\frac{\pi}{2}$ (V lags I)
Phasor diagram			
Variation of Z with *f*			
G,S_L,S_C (mho, seiman)	R does not depend on *f* G=1/R = conductance.	$S_L = 1/X_L$ Inductive susceptance	$S_C = 1/X_C$ Capacitive susceptance
Behaviour of device in D.C. and A.C	Same in A C and D C	L passes DC easily (because $X_L = 0$) while gives a high impedance for the A.C. of high frequency ($X_L \propto f$)	C - blocks DC (because $X_C = \infty$) while provides an easy path for the A.C. of high frequency $\left[X_C \propto \frac{1}{f}\right]$
Ohm's law	$V_R = IR$	$V_L = IX_L$	$V_C = IX_C$

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6. RESISTANCE AND INDUCTANCE IN SERIES (R-L CIRCUIT)

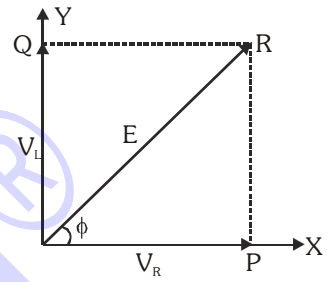
A circuit containing a series combination of a resistance R and an inductance L, connected with a source of alternating e.m.f. E as shown in figure.



6.1 Phasor diagram For L-r circuit

Let in a L-R series circuit, applied alternating emf is $E = E_0 \sin\omega t$. As R and L are joined in series, hence current flowing through both will be same at each instant. Let I be the current in the circuit at any instant and V_L and V_R the potential differences across L and R respectively at that instant.

Then $V_L = IX_L$ and $V_R = IR$



Now, V_R is in phase with the current while V_L leads the current by $\frac{\pi}{2}$.

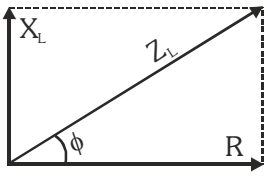
So V_R and V_L are mutually perpendicular (Note : $E \neq V_R + V_L$)

The vector OP represents V_R (which is in phase with I), while OQ represents V_L (which leads I by 90°).

The resultant of V_R and V_L = the magnitude of vector OR $E = \sqrt{V_R^2 + V_L^2}$

Thus $E^2 = V_R^2 + V_L^2 = I^2 (R^2 + X_L^2) \Rightarrow I = \dfrac{E}{\sqrt{R^2 + X_L^2}}$

The phasor diagram shown in fig. also shows that in L-R circuit the applied emf E leads the current I or conversely the current I lags behind the e.m.f. E. by a phase angle ϕ $\tan\phi = \dfrac{V_L}{V_R} = \dfrac{IX_L}{IR} = \dfrac{X_L}{R} = \dfrac{\omega L}{R}$



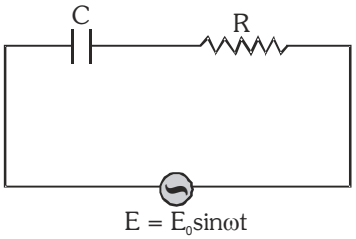
$\Rightarrow \phi = \tan^{-1}\left(\dfrac{\omega L}{R}\right)$

6.2 Inductive Impedance Z_L

In L-R circuit the maximum value of current $I_0 = \dfrac{E_0}{\sqrt{R^2 + \omega^2 L^2}}$ Here $\sqrt{R^2 + \omega^2 L^2}$ represents the effective opposition offered by L-R circuit to the flow of a.c. through it. It is known as impedance of L-R circuit and is represented by Z_L. $Z_L = \sqrt{R^2 + \omega^2 L^2} = \sqrt{R^2 + (2\pi fL)^2}$. The reciprocal of impedance is called admittance $Y_L = \dfrac{1}{Z_L} = \dfrac{1}{\sqrt{R^2 + \omega^2 L^2}}$

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7. RESISTANCE AND CAPACITOR IN SERIES (R-C CIRCUIT)



A circuit containing a series combination of a resistance R and a capacitor C, connected with a source of e.m.f. of peak value E_0 as shown in fig.

7.1 Phasor diagram For R-C circuit

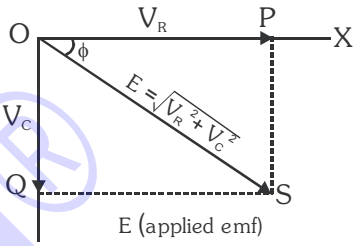
Current through both the resistance and capacitor will be same at every instant and the instantaneous potential differences across C and R are

$$V_C = I X_C \text{ and } V_R = I R$$

where X_C = capacitive reactance and I = instantaneous current.

Now, V_R is in phase with I, while V_C lags behind I by 90°.

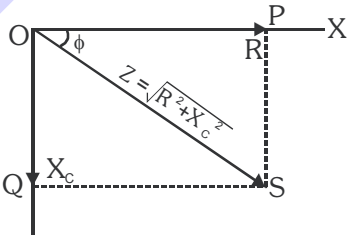
The phasor diagram is shown in fig.



The vector OP represents V_R (which is in phase with I)

and the vector OQ represents V_C (which lags behind I by $\frac{\pi}{2}$).

The vector OS represents the resultant of V_R and V_C = the applied e.m.f. E.



Hence $V_R^2 + V_C^2 = E^2 \Rightarrow E = \sqrt{V_R^2 + V_C^2}$

$$\Rightarrow E^2 = I^2 (R^2 + X_C^2) \Rightarrow I = \frac{E}{\sqrt{R^2 + X_C^2}}$$

The term $\sqrt{(R^2 + X_C^2)}$ represents the effective resistance of the R-C circuit and called the capacitive impedance Z_C of the circuit.

Hence, in C-R circuit $Z_C = \sqrt{R^2 + X_C^2} = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}$

7.2 Capacitive Impedance Z_C

In R-C circuit the term $\sqrt{R^2 + X_C^2}$ effective opposition offered by R-C circuit to the flow of a.c. through it. It is known as impedance of R-C circuit and is represented by Z_C

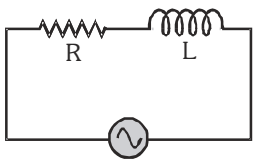
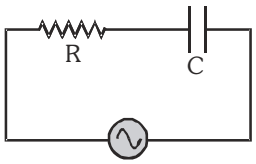
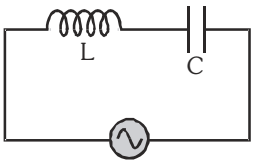
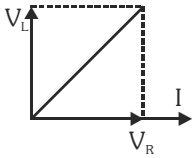
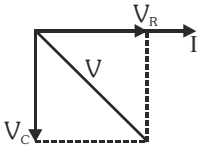
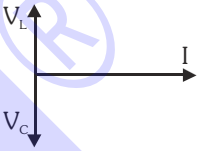
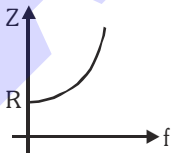
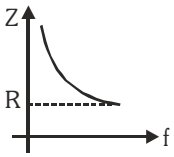
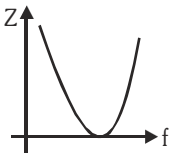
The phasor diagram also shows that in R-C circuit the applied e.m.f. lags behind the current I (or the current I leads the emf E) by a phase angle ϕ given by

$$\tan\phi = \frac{V_C}{V_R} = \frac{X_C}{R} = \frac{1/\omega C}{R} = \frac{1}{\omega CR}, \quad \tan\phi = \frac{X_C}{R} = \frac{1}{\omega CR} \Rightarrow \phi = \tan^{-1}\left(\frac{1}{\omega CR}\right)$$

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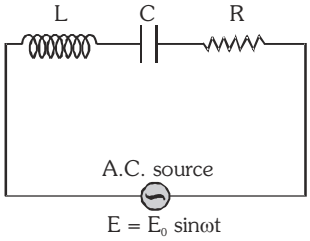
8. COMBINATION OF COMPONENTS (R-L or R-C or L-C)

TERM	R-L	R-C	L-C
Circuit			
	I is same in R & L	I is same in R & C	I is same in L & C
Phasor diagram			
	$V^2 = V_R^2 + V_L^2$	$V^2 = V_R^2 + V_C^2$	$V = V_L - V_C (V_L > V_C)$ $V = V_C - V_L (V_C > V_L)$
Phase difference	V leads I ($\phi = 0$ to $\frac{\pi}{2}$)	V lags I ($\phi = -\frac{\pi}{2}$ to 0)	V lags I ($\phi = -\frac{\pi}{2}$, if $X_C > X_L$) V leads I($\phi = +\frac{\pi}{2}$, if $X_L > X_C$)
Impedance	$Z = \sqrt{R^2 + X_L^2}$	$Z = \sqrt{R^2 + (X_C)^2}$	$Z = \lvert X_L - X_c \rvert$
Variation of Z	as f↑,Z ↑	as f↑,Z↓	as f↑, Z first ↓ then↑
with f			
At very low f	$Z \simeq R$ ($X_L \to 0$)	$Z \simeq X_C$	$Z \simeq X_C$
At very high f	$Z \simeq X_L$	$Z \simeq R$ ($X_C \to 0$)	$Z \simeq X_L$

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9. INDUCTANCE, CAPACITANCE AND RESISTANCE IN SERIES (L-C-R SERIES CIRCUIT)

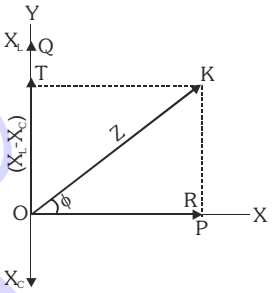
A circuit containing a series combination of an resistance R, a coil of inductance L and a capacitor of capacitance C, connected with a source of alternating e.m.f. of peak value of E_0, as shown in fig.



9.1 Phasor Diagram For Series L-C-R circuit

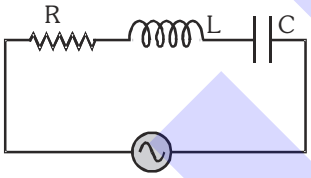
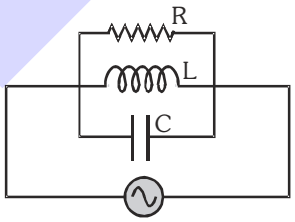
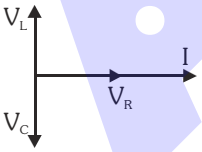
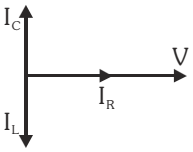
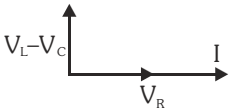
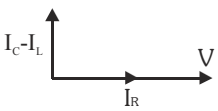
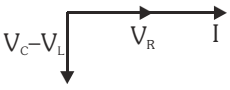
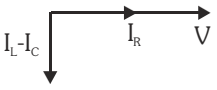
$$E=\sqrt{V_R^2+(V_L-V_C)^2}=I\sqrt{R^2+(X_L-X_C)^2}\Rightarrow I=\frac{E}{\sqrt{R^2+(X_L-X_C)^2}}$$

Impedance $Z=\sqrt{R^2+(X_L-X_C)^2}=\sqrt{R^2+\left(\omega L-\frac{1}{\omega C}\right)^2}$



The phasor diagram also shown that in LCR circuit the applied e.m.f. leads the current I by a phase angle ϕ $\quad \tan\phi=\frac{X_L-X_C}{R}$

10. SERIES LCR AND PARALLEL LCR COMBINATION

	SERIES L-C-R CIRCUIT	PARALLEL L-C-R CIRCUIT
1.	**Circuit diagram**	
		 V same for R,L and C
	I same for R, L & C	V same for R, L & C
2.	**Phasor diagram**	
		
(i)	If $V_L > V_C$ then	if $I_C > I_L$ then
		
(ii)	If $V_C > V_L$ then	if $I_L > I_C$ then
		

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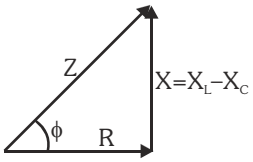
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(iii) $V = \sqrt{V_R^2 + (V_L - V_C)^2}$

Impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$

$\tan\phi = \dfrac{X_L - X_C}{R} = \dfrac{V_L - V_C}{V_R}$

(iv) Impedance triangle

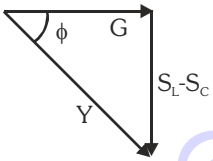


(iii) $I = \sqrt{I_R^2 + (I_L - I_C)^2}$

Admittance $Y = \sqrt{G^2 + (S_L - S_C)^2}$

$\tan\phi = \dfrac{S_L - S_C}{G} = \dfrac{I_L - I_C}{I_R}$

(iv) Admittance triangle



Series

(a) if $X_L > X_C$ then V leads I,ϕ (positive) circuit nature inductive

(b) if $X_C > X_L$ then V lags I,ϕ (negative) circuit nature capacitive

Parallel

(a) if $S_L > S_C$ ($X_L < X_C$) then V leads I,ϕ (positive) circuit nature inductive

(b) if $S_C > S_L$ ($X_C < X_L$) then V lags I, ϕ (negative) circuit nature capacitive

- In A.C. circuit voltage for L or C may be greater than source voltage or current but it happens only when circuit contains L and C both and on R it never greater than source voltage or current.
- In parallel A.C.circuit phase difference between I_L and I_C is π

11. RESONANCE

There are two types of resonance : (i) Series Resonance (ii) Parallel Resonance

11.1 Series Resonance

(a) At Resonance

(i) $X_L = X_C$ (ii) $V_L = V_C$ (iii) $\phi = 0$ (V and I in same phase)

(iv) $Z_{min} = R$ (impedance minimum) (v) $I_{max} = \dfrac{V}{R}$ (current maximum)

(b) Resonance frequency

$$\because \quad X_L = X_C \quad \Rightarrow \quad \omega_r L = \frac{1}{\omega_r C} \quad \Rightarrow \quad \omega_r^2 = \frac{1}{LC} \quad \Rightarrow \omega_r = \frac{1}{\sqrt{LC}} \Rightarrow f_r = \frac{1}{2\pi\sqrt{LC}}$$

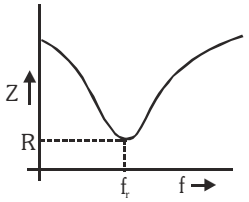
(c) Variation of Z with f

(i) If $f < f_r$ then $X_L < X_C$ circuit nature capacitive, ϕ (negative)

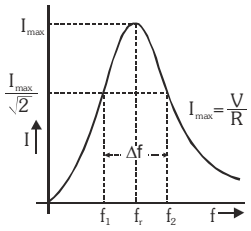
(ii) At $f = f_r$ then $X_L = X_C$ circuit nature, Resistive, ϕ = zero

(iii) If $f > f_r$ then $X_L > X_C$ circuit nature is inductive,ϕ (positive)

Variation of I with f as f increase, Z first decreases then increase



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(d) 

as f increase, I first increase then decreases

- At resonance impedance of the series resonant circuit is minimum so it is called 'acceptor circuit' as it most readily accepts that current out of many currents whose frequency is equal to its natural frequency. In radio or TV tuning we receive the desired station by making the frequency of the circuit equal to that of the desired station.

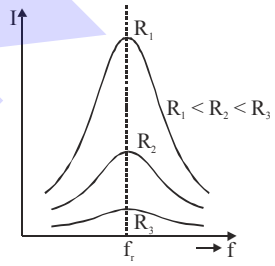
Half power frequencies

The frequencies at which, power become half of its maximum value called half power frequencies

Band width $= \Delta f = f_2 - f_1$

Quality factor Q : Q-factor of AC circuit basically gives an idea about stored energy & lost energy.

$$Q = 2\pi \frac{\text{maximum energy stored per cycle}}{\text{maximum energy loss per cycle}}$$



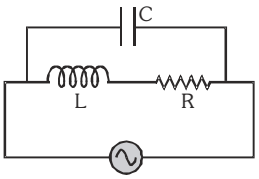
(i) It represents the sharpness of resonance.

(ii) It is unit less and dimension less quantity

(iii) $Q = \frac{(X_L)_r}{R} = \frac{(X_C)_r}{R} = \frac{2\pi f_r L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{f_r}{\Delta f} = \frac{f_r}{\text{band width}}$

11.2 Parallel Resonance

(a) At resonance



(i) $S_L = S_C$

(ii) $I_L = I_C$

(iii) $\phi = 0$

(iv) $Z_{max} = R$ (impedance maximum)

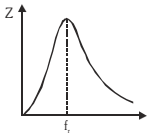
(v) $I_{min} = \frac{V}{R}$ (current minimum)

(b) Resonant frequency $f_r = \frac{1}{2\pi\sqrt{LC}}$

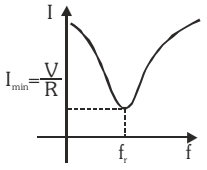
(c) Variation of Z with f as f increases, Z first increases then decreases

If $f < f_r$ then $S_L > S_C$, ϕ (positive), circuit nature is inductive

If $f > f_r$ then $S_C > S_L$, ϕ (negative), circuit nature capacitive.



(d) Variation of I with f as f increases, I first decreases then increases



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12. POWER IN AC CIRCUIT

The average power dissipation in LCR AC circuit

Let $V = V_0 \sin\omega t$ and $I = I_0 \sin(\omega t - \phi)$

Instantaneous power $P = (V_0 \sin\omega t)(I_0 \sin(\omega t - \phi) = V_0 I_0 \sin\omega t (\sin\omega t\cos\phi - \sin\phi\cos\omega t)$

$$\text{Average power } \langle P\rangle = \frac{1}{T}\int_0^T (V_0 I_0 \sin^2\omega t\cos\phi - V_0 I_0 \sin\omega t\cos\omega t\sin\phi)dt$$

$$= V_0 I_0\left[\frac{1}{T}\int_0^T \sin^2\omega t\cos\phi dt - \frac{1}{T}\int_0^T \sin\omega t\cos\omega t\sin\phi dt\right] = V_0 I_0\left[\frac{1}{2}\cos\phi - 0\times\sin\phi\right]$$

$$\Rightarrow \quad \langle P\rangle = \frac{V_0 I_0\cos\phi}{2} = V_{rms}\, I_{rm,s}\cos\phi$$

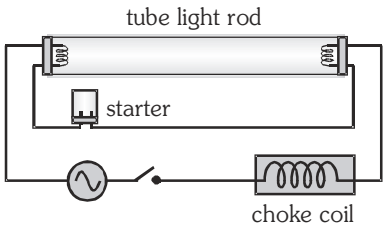
- $I_{rms}\cos\phi$ is known as active part of current or wattfull current, workfull current. It is in phase with voltage.
- $I_{rms}\sin\phi$ is known as inactive part of current, wattless current, workless current. It is in quadrature (90°) with voltage.

Pure/Ideal	ϕ	V	Power factor = $\cos\phi$	Average power
R	0	V, I same Phase	1 (maximum)	$V_{rms}.\ I_{rms}$
L	$+\frac{\pi}{2}$	V leads I	0	0
C	$-\frac{\pi}{2}$	V lags I	0	0

- At resonance power factor is maximum ($\phi = 0$ so $\cos\phi = 1$) and $P_{av} = V_{rms}\, I_{rms}$

13. CHOKE COIL

In a direct current circuit, current is reduced with the help of a resistance. Hence there is a loss of electrical energy $I^2 R$ per sec in the form of heat in the resistance. But in an AC circuit the current can be reduced by choke coil which involves very small amount of loss of energy. Choke coil is a copper coil wound over a soft iron laminated core. This coil is put in series with the circuit in which current is to be reduced. It also known as ballast.



Circuit with a choke coil is a series L-R circuit. If resistance of choke coil = r (very small)

The current in the circuit $I = \frac{E}{Z}$ with $Z = \sqrt{(R+r)^2 + (\omega L)^2}$ So due to large inductance L of the coil, the current in the circuit is decreased appreciably. However, due to small resistance of the coil r,

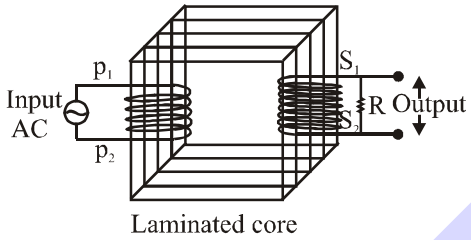
The power loss in the choke $P_{av} = V_{rms}\, I_{rms}\cos\phi \to 0 \ \because\ \cos\phi = \frac{r}{Z} = \frac{r}{\sqrt{r^2 + \omega^2 L^2}} \approx \frac{r}{\omega L} \to 0$

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14. TRANSFORMERS

One of the great advantages of ac over dc for electric-power distributi on is that it is much easier to step voltage levels up and down with ac than with dc. The necessary conversion is accomplished by a static device called transformer using the principle of mutual induction.

The figure shows an idealised transformer which consists of two coils or windings, electrically insulated from each other but wound on the same core. The winding to which power is supplied is called primary, the winding from which power is delivered is called the secondary.



The ac source causes an alternating current in the primary which sets up an alternating flux in the core and this induces an emf in each winding of secondary in accordance with Faraday's law. For ideal transformer we assume that primary has negligible resistance and all the flux in core links both primary and secondary.

The primary winding has N_1 turns and secondary has N_2 turns. When the magnetic flux changes because of changing currents in the two coils, the resulting induced emf are

$$E_1 = -N_1 \frac{d\phi_B}{dt} \text{ and } E_2 = -N_2 \frac{d\phi_B}{dt}$$

The flux per turn B is same in both primary and the secondary so that the emf per turn is same in each. The ratio of secondary emf E_2 to the primary emf E_1 is therefore equal at any instant to the ratio of secondary to primary turns.

$$\frac{E_2}{E_1} = \frac{N_2}{N_1}$$

If the windings have zero resistance, the induced emf 1 and 2 are equal to the terminal voltage across the primary and the secondary respectively, hence

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

If $N_2 > N_1$ then $V_2 > V_1$ and we have step up transformer, if $N_2 < N_1$ then $V_2 < V_1$ and we have a step down transformer.

If the transformer is assumed to be 100% efficient (no energy losses) the power input is equal to the power output i.e.

$$I_2 V_2 = I_1 V_1 \qquad \therefore \qquad \frac{I_2}{I_1} = \frac{N_1}{N_2}$$

All the currents and voltages derived above have same frequency as that of source. The equations obtained above apply to ideal transformers, although some energy is lost but well designed transformers have efficiency more than 95%, this is a good approximation. The causes of energy losses and their rectification is given below

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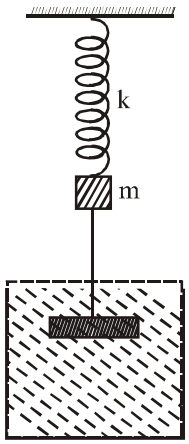
	Cause		Rectification
1	Due to poor design and air gaps in The core, all the flux due to primary does not pass through the secondary.	1	By winding the primary and secondary coil one over the other.
2	Resistance of windings causes I^2R loss.	2	In high current, low voltage, these are minimised by using thick wire.
3	The alternating magnetic flux induces eddy currents in the core and causes heating.	3	By using laminated core it can be reduced.
4	Alternating magnetisating of core causes hystersis loss.	4	It is kept minimum by using a magnetic material having low hystersis loss. (e.g. soft iron)

15. DAMPED HARMONIC OSCILLATOR

In the previous section we have considered several examples of oscillatory systems. In each case the amplitude of the oscillation does not depend on time and the system, once set to oscillate, will continue to do so forever. Physical systems are never as ideal as that. Even in the most controlled case there will be dissipative elements.

In each of these cases the amplitudes of the oscillation will gradually decrease and the motion will be 'damped'.

Most of the time we are interested in the motion of an object through air or other viscous fluids. The motion of the object is then subject to a resistive force called aerodynamic or viscous drag. Experimentally the resistive force is found to be proportional to the velocity of the body for low relative speed with respect to the medium.



A laboratory model for a damped spring-mass system is shown in figure. Where a vertical spring has at its end a mass m which is connected to massless piston which moves through a liquid. For low speeds it is reasonable to write the frictional force as

$$F_f = -\alpha v \qquad (\alpha \text{ in kg/s})$$

where α is a positive constant known as damping constant. The equation of motion of a harmonic oscillator becomes

$$m\frac{d^2x}{dt^2} = -kx - \alpha\frac{dx}{dt}$$

which can be rewritten as

$$\frac{d^2x}{dt^2} + 2\gamma\frac{dx}{dt} + \omega_0^2 x = 0$$

where $2\gamma = \alpha/m > 0$ and ω_0^2, as before is equal to k/m. γ is known as damping coefficient

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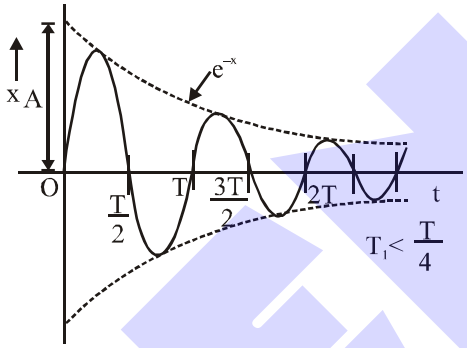
15.1 Damped Oscillations (Underdamped oscillation)

If the dissipative forces are small, i.e. if $\gamma^2 < \omega_0^2$, the dominant force acting on the mass is the conservation restoring force. The system therefore shows oscillations, even though the amplitudes of oscillation continuously decrease with time.

we get $x = Ae^{-\gamma t}\sin(\Omega t + \phi)$

where the new angular frequency of oscillation Ω is given by $\Omega^2 = \omega_0^2 - \gamma^2$

i.e the frequency shift to a lower value due to damping. The amplitude of oscillation is no longer constant but decreases with time in an exponential fashion. The variation of x with time is shown in figure.



It may be seen that sin ($\Omega t + \phi$) is a periodic function with period $2\pi/\Omega$. when this is multiplied with $e^{-\gamma t}$ the zeroes of the product would still be at the same place where sin ($\Omega t + \phi$) becomes zero. The maxima however does not fall midway between the two minima because of the exponential $e^{-\gamma t}$ but occurs at a slightly earlier time.

Power Loss in a Damped Oscillation-Q-factor

Because of the work done against the nonconservative forces, the energy stored in an oscillator continuously decreases. The fraction of energy lost in a period with respect to its average value for that period is a measure of the quality of the oscillator. The less the value of this fraction is, the more is the ability of the oscillator to sustain periodic motion. Quantitatively, the quality of an oscillator is measured through a Q-factor defined as

$$Q = 2\pi \times \frac{\text{Average energy stored in one period}}{\text{average loss of energy in that period}}$$

If the damping is light i.e. $\gamma << \omega_0$, The instantaneous energy of the oscillator is

$$E(t) = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

$$= \frac{1}{2}mA^2\omega_0^2 e^{-2\gamma t}$$

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where we have replaced k by $m\omega_0^2$. At t = 0 the energy of the oscillator is $E_0 = mA^2\,\omega_0^2/2$ so that

$$E(t) = E_0 e^{-2\gamma t}$$

$$P(t) = \frac{dE}{dt} = -2\gamma E_0 e^{-2\gamma t} = 2\gamma\, E(t)$$

Energy lost in one period can be approximately written as

$$< p(t) > \frac{2\pi}{\Omega} = 2\gamma < E(t) > \frac{2\pi}{\Omega}$$

where the angular bracket < ..... > denotes the time average value during a period at time t. The quality or Q-factor for an oscillator is

$$2\pi \frac{< E(t) >}{2\gamma < E(t) > \frac{2\pi}{\Omega}} = \frac{\Omega}{2\pi} \cong \frac{\omega_0}{2\gamma}$$

It may be noted that Q is a dimensionless quantity, greater than unity by our assumption that $\gamma << \omega_0$.

Time to reach extreme

The natural logarithm of two successive maxima is called logarithmic decrement and is denoted by λ. From Equation it follows that

$$\lambda = \frac{2\pi\gamma}{\Omega} = \frac{\pi}{Q}$$

15.2 Overdamped Motion

In equation if $\gamma > \omega_0$ we do not get oscillatory solution

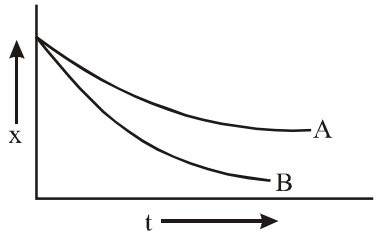
15.3 Critical Damping ($\gamma = \omega_0$)

The solution for the displacement is therefore

$$x = (A + Bt)e^{-\gamma t}$$

$$x' = -\frac{mg}{k}(1 + \gamma t)e^{-\gamma t}$$

It may be seen (hat once again there are no oscillation. As $t \to \infty$, $x \to 0$ so that the particle takes an infinite time before reaching the equilibrium position. The difference with the overdamped case is that the approach to the equilibrium is faster. The variation of displacement with time is shown in figure (curve B.)



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16. FORCED OSCILLATIONS AND RESONANCE

The oscillations in a system can be indefinitely maintained by supplying energy continuously. In mechanical system this can be done by subjecting it to an external force which itself has harmonic time dependence.

An interesting phenomenon occurs if the frequency of the external source is equal or nearly equal to the natural frequency of the system. The amplitude of oscillations is found to increase many folds in such cases. This is called resonance.

Forced Damped Oscillations

In the presence of resistive forces proportional to velocity, equation be comes

$$\frac{d^2x}{dt^2} + 2\gamma\frac{dx}{dt} + \omega_0^2 x = \frac{F_0}{m}\cos\omega t$$

Let us therefore try a solution of the type

$x = A\cos\omega t + B\sin\omega t + Ce^{-\lambda t}$

The steady - state displacement is then given by

$x = C\cos(\omega t + \delta)$

with

$$C = \frac{F_0}{m}\sqrt{\frac{1}{\left(\omega_0^2 - \omega^2\right)^2 + 4\omega^2\gamma^2}}$$

and $\tan\delta = \dfrac{2\omega\gamma}{\omega^2 - \omega_0^2}$

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